Research development summary

Trace Sparsity and Arithmeticity for Compact Hyperbolic Surfaces

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This abridged retrospective describes the research task and mathematical route recorded in the project's released documentation. The brief below is a schematic editorial summary; it is not a quotation of an original prompt or a reconstruction of a conversation.

1 Research brief

The task was to investigate whether sparse signed traces force a cocompact Fuchsian lattice to be arithmetic, connecting the problem to the compact Sarnak bounded-clustering and Schmutz linear-growth conjectures. The research brief can be summarized as follows:

- identify which trace-growth hypotheses suffice for arithmeticity;
- test candidate approaches and make their limitations explicit;
- develop the required estimates with their external hypotheses and uniformity ranges stated; and
- consolidate the argument into a focused manuscript and check suitable algebraic and logical components in Lean.

The manuscript ultimately presents the stronger critical-growth statement. Writing $N_\Gamma(X)$ for the number of distinct signed traces in $[-X, X]$, its main claim is

$$\limsup_{X\to\infty} \frac{\log N_\Gamma(X)}{\log X} \le 1 \quad \Longrightarrow \quad \Gamma \text{ is arithmetic.}$$

Bounded clustering implies linear growth by an interval-covering argument, and linear growth implies this critical bound. The named compact conjectures follow as consequences.

2 Schematic development

The project records describe an iterative process of formulating approaches, testing their mathematical sufficiency, isolating estimates, and assembling a proof. Its main stages were:

1. **Set up the arithmetic framework.** Work with signed traces, the subgroup generated by squares, the invariant trace field, and trace-of-powers recurrences. These connect hyperbolic geometry to arithmetic criteria.

2. **Identify the limits of preliminary approaches.** Close trace pairs alone do not contradict bounded clustering. Scalar or additive arguments and a universal radical estimate for one fixed recurrence did not supply the needed general conclusion.

3. **Change which parameter varies.** Fix the index in the Chebyshev recurrence and vary the group element. This is the recorded strategic change that makes known bounded-power expansion results applicable to the required recurrence moduli.

4. **Combine arithmetic and counting estimates.** Use a Roth-type divisor lower bound, a prime-square selection argument, uniform two-trace fiber estimates, and a congruence count on a trace graph. State the dependence of constants and the order of limits.

5. **Assemble and check the final argument.** Deduce arithmeticity using local expansion, algebraic specialization, and rigidity. Check the cited interfaces, the exposition, and selected identities and conditional implications through the Lean package.

3 The resulting mathematical mechanism

The central estimate compares an arithmetic local-expansion functional at an algebraic specialization y with geometric length at the original Fuchsian character x:

$$\mathcal{J}_{k_y}(\operatorname{tr}_y b) \leq \frac{\ell_x(b)}{2}.$$

The moving-element construction provides the arithmetic lower estimate and a compatible counting upper estimate. Passing to the limits in the stated order yields the factor $1/2$.

For an algebraic character, the distinguished real place already contributes this half-length. The remaining nonnegative local terms must vanish, giving the trace conditions used in Takeuchi's arithmeticity criterion. For a transcendental character, algebraic specialization and the estimate give marked-length domination; Thurston's separating asymmetric metric is then used to force equality of the characters. A finite-index argument extends the compact result to orbifolds. The manuscript contains the precise statements and proofs.

4 Verification and provenance

The supporting Lean project checks selected recurrence and polynomial identities, finite combinatorial inequalities, limiting coefficient arguments, and deductions from explicit analytic, geometric, and arithmetic hypotheses. The full geometric arithmeticity argument and the proofs supplying those external inputs remain in the manuscript.

This account is based on released project notes and audit records. No original conversation export was available for authentication, so the summary makes no claim about exact prompt wording or individual model turns. Source coverage and verification details are documented separately. Account handles, account and session identifiers, dialogue logs, local paths, and operational details are omitted. The contributor names and affiliations on the first page are intentional public attribution.